All India Aakash Test Series for NEET - 2027

TEST - I (Code-E)

Test Date : 01/11/2026

ANSWERS

1. (3)	37. (4)	73. (3)	109. (3)	145. (2)
2. (1)	38. (1)	74. (2)	110. (4)	146. (4)
3. (4)	39. (2)	75. (4)	111. (2)	147. (3)
4. (1)	40. (3)	76. (2)	112. (3)	148. (2)
5. (1)	41. (4)	77. (1)	113. (4)	149. (4)
6. (2)	42. (2)	78. (4)	114. (2)	150. (3)
7. (2)	43. (4)	79. (1)	115. (3)	151. (2)
8. (4)	44. (4)	80. (2)	116. (4)	152. (3)
9. (2)	45. (1)	81. (2)	117. (2)	153. (2)
10. (4)	46. (3)	82. (2)	118. (1)	154. (3)
11. (1)	47. (4)	83. (3)	119. (3)	155. (4)
12. (3)	48. (1)	84. (1)	120. (2)	156. (2)
13. (4)	49. (3)	85. (3)	121. (2)	157. (3)
14. (3)	50. (1)	86. (1)	122. (1)	158. (2)
15. (3)	51. (2)	87. (2)	123. (3)	159. (3)
16. (1)	52. (4)	88. (2)	124. (1)	160. (2)
17. (4)	53. (4)	89. (1)	125. (2)	161. (4)
18. (3)	54. (1)	90. (3)	126. (1)	162. (2)
19. (1)	55. (2)	91. (2)	127. (3)	163. (2)
20. (1)	56. (2)	92. (2)	128. (2)	164. (2)
21. (2)	57. (1)	93. (1)	129. (1)	165. (3)
22. (3)	58. (4)	94. (3)	130. (1)	166. (2)
23. (3)	59. (4)	95. (2)	131. (3)	167. (3)
24. (3)	60. (1)	96. (2)	132. (1)	168. (3)
25. (1)	61. (1)	97. (1)	133. (3)	169. (2)
26. (2)	62. (3)	98. (1)	134. (2)	170. (1)
27. (4)	63. (1)	99. (3)	135. (1)	171. (4)
28. (3)	64. (3)	100. (2)	136. (2)	172. (3)
29. (3)	65. (1)	101. (4)	137. (3)	173. (3)
30. (4)	66. (2)	102. (4)	138. (1)	174. (4)
31. (2)	67. (4)	103. (2)	139. (4)	175. (4)
32. (2)	68. (4)	104. (3)	140. (3)	176. (3)
33. (3)	69. (3)	105. (3)	141. (1)	177. (3)
34. (4)	70. (1)	106. (2)	142. (2)	178. (2)
35. (3)	71. (4)	107. (3)	143. (3)	179. (3)
36. (3)	72. (4)	108. (2)	144. (4)	180. (2)

HINTS & SOLUTIONS

[PHYSICS]

1. Answer (3)

Hint: We know that energy stored in capacitor $= \frac{1}{2}CV^2$.

Hence, CV^2 denotes energy

Sol.: $[CV^2] \rightarrow$ Energy $\rightarrow [M^1L^2T^{-2}]$

$\therefore a + b + c = 1 + 2 - 2 = 1$

2. Answer (1)

Hint: Least count $= \frac{\text{Pitch}}{\text{Number of division on circular scale}}$

Sol.: Least count $= \frac{0.5 \text{ mm}}{50} = 0.01$ mm

3. Answer (4)

Hint: $\omega = \frac{d\theta}{dt}$ and $\alpha = \frac{d\omega}{dt}$

Sol.: $\omega = \frac{d\theta}{dt} = \frac{d}{dt}\left[\frac{5t^3}{30} - \frac{t^2}{2}\right]$

$= \frac{5 \times 3t^2}{30} - \frac{2t}{2}$

$\omega = \frac{t^2}{2} - t$

$\alpha = \frac{d\omega}{dt} = \frac{2t}{2} - 1 = t - 1 = 10 - 1$

$= 9$ rad/s^2

4. Answer (1)

Hint: $a_x = v_x \frac{dv_x}{dx}; a_y = v_y \frac{dv_y}{dy}$ and $a = \sqrt{a_x^2 + a_y^2}$

Sol.: $a_x = -x\frac{d(-x)}{dx}$ and $a_y = 2y\frac{d}{dy}(2y)$

$a_x = x$ and $a_y = 2y \times 2 = 4y$

$a = \sqrt{a_x^2 + a_y^2} = \sqrt{x^2 + (4y)^2}$

at point (1, 2) m

$a = \sqrt{1^2 + (4 \times 2)^2} = \sqrt{65}$ m/s^2

5. Answer (1)

Hint: Horizontal range of a projectile is given by $R_\theta = \frac{u^2 \sin 2\theta}{g}$

Sol.: Here u and g are same

$\therefore \quad R \propto \sin 2\theta$

$\Rightarrow \frac{R_{30^\circ}}{R_{60^\circ}} = \frac{\sin 60^\circ}{\sin 120^\circ} = \frac{\frac{\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{1}$

6. Answer (2)

Hint: Use velocity, $v = \frac{dx}{dt}$

Sol.: $\frac{dx}{dt} = 2t \Rightarrow dx = 2tdt$

Integrating both sides,

$\int_0^x dx = 2\int_0^2 tdt$

$\Rightarrow x - 0 = 2\left[\frac{t^2}{2}\right]_0^2$

$\Rightarrow x = 2^2 = 4$ m

7. Answer (2)

Hint & Sol.: [Angular Frequency] $= \left[\frac{2\pi}{T}\right] = [M^0L^0T^{-1}]$

[Planck's constant h] $= \left[\frac{E}{\nu}\right] = \frac{[ML^2T^{-2}]}{\left[\frac{1}{T}\right]}$

$= [ML^2T^{-1}]$

[Torque] = [Energy] = $[ML^2T^{-2}]$

[Electric potential] $= \left[\frac{W}{Q}\right] = \left[\frac{ML^2T^{-2}}{AT}\right]$

$= [ML^2T^{-3}A^{-1}]$

8. Answer (4)

Hint: Speed of stone hitting the car B is relative speed of stone w.r.t. car B, and similarly speed v of stone will be speed of stone w.r.t. car A.

Sol.: $(v_{SB}) = |v_S - v_B|$

$\Rightarrow 10 \times \frac{18}{5} = v_S - 80$

$\Rightarrow 36 + 80 = v_S \Rightarrow v_S = 116$ km/h

Now $v_{SA} = |v_S - v_A|$

$v = v_S - v_A = 116 - 100$

$v = 16$ km/h

9. Answer (2)

Hint: Use $\tan\alpha = \frac{V_y}{V_x}$

Sol.: $V_x = \frac{dx}{dt} = 20, V_y = \frac{dy}{dt} = 50 - 10t$

at $t = 2$ s,

$V_x = 20$ m/s and $V_y = 50 - 20 = 30$ m/s

$\tan\alpha = \frac{V_y}{V_x} = \frac{30}{20} = \frac{3}{2}$

10. Answer (4)

Hint and Sol: At complementary angles of projection, horizontal ranges are same while their time of flights are different.

$\therefore$ Average velocities are different.

11. Answer (1)

Hint: $R = \frac{V}{I}$ and $\left(\frac{\Delta R}{R}\right) = \frac{\Delta V}{V} + \frac{\Delta I}{I}$

Sol.: $R = \frac{10}{5} = 2\ \Omega$

and $\frac{\Delta R}{R} = \frac{0.1}{10} + \frac{0.2}{5} = \frac{0.5}{10}$

$(\Delta R)_{max} = \frac{0.5}{10} \times 2 = \frac{1}{10} = 0.1\ \Omega$

12. Answer (3)

Hint: Use Galileo's law of odd numbers

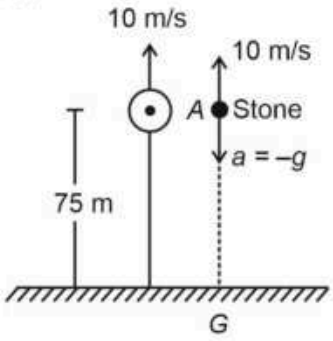
Sol.: here $\frac{\theta_1}{\theta_2} = \frac{1}{3}$

So $\frac{\theta_2}{3\theta_1} = \frac{3}{3} = 1$

13. Answer (4)

Hint: Find time taken by stone to reach the ground and then find separation between stone and balloon.

Sol.:



Now using equation of motion $S = ut + \frac{1}{2}at^2$ from A to G.

$\Rightarrow -75 = 10t - \frac{1}{2}10t^2$

$-75 = 10t - 5t^2$

$\Rightarrow -15 = 2t - t^2$

$\Rightarrow t^2 - 2t - 15 = 0$

$\Rightarrow (t - 5)(t + 3) = 0$

$\Rightarrow t = 5$ s and $t = -3$ s (Not possible)

So, height of balloon at this instant

$= 75 + 10 \times 5 = 125$ m

14. Answer (3)

Hint: Use distance = speed × time

Sol.: Distance = $\alpha \times t = \alpha \times (12 \times 60 + 30)$

$= 750\alpha$

15. Answer (3)

Hint: Use average velocity, $V_{avg} = \frac{x_1 + x_2}{\frac{x_1}{v_1} + \frac{x_2}{v_2}}$

Sol.: $20 = \frac{\frac{x}{2} + \frac{3x}{2}}{\frac{x}{2 \times 10} + \frac{3x}{2 \times v_2}}$

$\Rightarrow 20 = \frac{2x}{\frac{x}{20} + \frac{3x}{2v_2}} = \frac{2}{\frac{1}{20} + \frac{3}{2v_2}}$

$\Rightarrow \frac{1}{2} + \frac{15}{v_2} = 1 \Rightarrow v_2 = 30$ m/s

16. Answer (1)

Hint and sol.: Zero between two non-zero digits is significant and when number has decimal point then trailing zeros are significant. Hence number of significant zeros are two.

17. Answer (4)

Hint & Sol: A and B have different dimensions so they cannot be added or subtracted, hence option (4) $Z = \dfrac{AB^2}{A+B}$ is certainly not meaningful.

18. Answer (3)

Hint & Sol: Precision depends upon least count of instrument.

19. Answer (1)

Hint & Sol:

- Random error can be minimised by taking more number of readings.
- Number of reliable digits plus one uncertain digit are known as significant figures in a measurement.

20. Answer (1)

Hint : Use application of dimensional analysis.

Sol: Let $F = [\rho]^a [f]^b [v]^c$

Writing dimension of both sides

$[MLT^{-2}] = [ML^{-3}]^a [T^{-1}]^b [LT^{-1}]^c$

equating power of M, L and T on both sides

$a = 1, -3a + c = 1$ and $-b - c = -2$

solving these we have

$a = 1, b = -2$ and $c = 4$

So, $F = [\rho^1 f^{-2} v^4]$

21. Answer (2)

Hint & sol.: Area = $l \times b = 1.62 \times 2.1$

$= 3.402$

$= 3.4\ m^2$

22. Answer (3)

Hint: Use $n_1u_2 = n_2u_2$

Sol.: $1\ kg\ m^2\ s^{-2} = x(10\ g)\ (1\ m)^2 \times (10\ s)^{-2}$

$\Rightarrow 1\ kg\ m^2\ s^{-2} = x(10)(10^{-2}) kg\ m^2\ s^{-2}$

$\Rightarrow 1 = 10^{-1}x$

$\Rightarrow x = 10$

23. Answer (3)

Hint: Use average acceleration, $a_{avg} = \left|\dfrac{\Delta\vec{v}}{\Delta t}\right|$

Sol.:

$$a_{avg} = \left|\frac{\vec{v}_2 - \vec{v}_1}{\Delta t}\right| = \frac{v\sqrt{2}}{\pi r / 2v} = \frac{2\sqrt{2}}{\pi \times \frac{4}{4}} = \frac{2\sqrt{2}}{\pi}\ m/s^2$$

24. Answer (3)

Hint: For complementary angles of projection, ranges are same for the same speed of projection.

Sol.: $t_1 = \dfrac{2u\sin\theta}{g}$ and $t_2 = \dfrac{2u\cos\theta}{g}$

$$t_1^2 + t_2^2 = \frac{4u^2}{g^2}\sin^2\theta + \frac{4u^2\cos^2\theta}{g^2}$$

$$t_1^2 + t_2^2 = \frac{4u^2}{g^2}$$

$$u^2 = \frac{g^2}{4}\left(t_1^2 + t_2^2\right)$$

$$u = \frac{1}{2}g\left(t_1^2 + t_2^2\right)^{1/2}$$

25. Answer (1)

Hint & Sol.: $\dfrac{d\vec{v}}{dt}$ is net acceleration of particle.

$\left|\dfrac{d\vec{v}}{dt}\right|$ is magnitude of net acceleration of particle

$\dfrac{d|\vec{v}|}{dt}$ is tangential acceleration of particle.

26. Answer (2)

Hint: Horizontal distance travelled by both object and helicopter will be same.

Sol.: Displacement of object w.r.t helicopter, will be vertically downward *i.e.*, S_{OH} = 2 km (↓)

27. Answer (4)

Hint & Sol.: Given time, t = 100 s

So, displacement of trolley = 5 × 100 = 500 m

Displacement of boy w.r.t. trolley is zero (because he returns to the same position w.r.t. trolley).

∴ Displacement of boy w.r.t. ground = 500 m

28. Answer (3)

Hint: Use symmetry of projectile motion, time taken by projectile in upward journey equals to downward journey and total time of flight $T = 2\sqrt{\dfrac{2h_{max}}{g}}$

Sol.: $t_{OA} = 2\sqrt{2\dfrac{H}{g}} = 2s$ ...(1)

and $t_{PQ} = 2\sqrt{\dfrac{2\times H/2}{g}} = 2\sqrt{H/g}$ ...(2)

From equation (1) and (2)

$$\frac{t_{PQ}}{t_{OA}} = \frac{1}{\sqrt{2}}$$

$$t_{PQ} = \frac{t_{OA}}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}\ s$$

29. Answer (3)

Hint: Maximum and minimum resultant of two vectors $\vec{A}$ and $\vec{B}$ are $(A + B)$ and $(A - B)$ respectively

Sol.: $A + B = 5$ ...(1)

and $A - B = 3$ ...(2)

Solving equation (1) and (2), we get

$A = 4$ and $B = 1$

Therefore $AB = 4$

30. Answer (4)

Hint & Sol.: When a projectile reaches at highest point of its trajectory in oblique projectile motion, then its speed and kinetic energy are minimum (but not zero). Acceleration of projectile is constant and it is equal to g at every point.

31. Answer (2)

Hint & Sol.: A car in uniform motion covers equal distances in equal intervals of time along a straight line with constant velocity.

$\therefore$ The x-t graph will have constant slope.

32. Answer (2)

Hint & Sol.: $\omega = 2\pi\nu = \dfrac{2\times 3.14\times 100}{60}$

= 10.47 rad/s

33. Answer (3)

Hint: Time taken to hit the ground, $T = \sqrt{\dfrac{2H}{g}}$

Sol.: $T = \sqrt{\dfrac{2\times 19.6}{9.8}} = \sqrt{4} = 2$ s

$\therefore v = u + gT = 0 + 9.8 \times 2 = 19.6$ m/s

34. Answer (4)

Hint & Sol: If the speed of body is constant all curved paths are possible; while in accelerated motion body can have constant speed, for example uniform circular motion.

35. Answer (3)

Hint & Sol: In uniform circular motion,

Angular velocity, $\omega \rightarrow$ is constant

$\Rightarrow \omega = \dfrac{d\theta}{dt}$

$\Rightarrow \int_0^{\theta} d\theta = \int_0^{t} \omega.dt$

$\theta = \omega t$

and as we know in uniform circular motion

$a = a_c = \dfrac{v^2}{r} \rightarrow$ non-zero

and $\vec{v} \rightarrow$ variable

$\because$ Direction of $\vec{v}$ is changing continuously.

36. Answer (3)

Hint & Sol.: Net displacement $= \sqrt{(12)^2 + (16)^2}$

$= \sqrt{400} = 20$ m

37. Answer (4)

Hint: Least count of vernier callipers,

LC = 1 MSD – 1 VSD

Sol.: LC = 1 MSD – $\dfrac{9}{10}$ MSD

$= \left(1 - \dfrac{9}{10}\right) 0.1$ cm

$= \dfrac{0.1}{10} = 0.01$ cm

38. Answer (1)

Hint & Sol.:

- Mass, time and velocity can be fundamental quantities in a system of units.
- Dimensionless quantities can have units.
- Unitless quantity exists.

39. Answer (2)

Hint: For dimensionally correct equation, each term must have same dimensions.

Sol.: $[F] = [Pt^{-1}]$

$\Rightarrow [P] = \left[\dfrac{F}{t^{-1}}\right] = [F \cdot t]$

$\therefore [P] = [MLT^{-2}] \times [T] = [MLT^{-1}]$

= [Momentum]

40. Answer (3)

Hint: Use equation of kinematics of motion.

Sol.: $s = \dfrac{1}{2}at^2 \Rightarrow t = \sqrt{\dfrac{2s}{a}}$

Time taken for covering 1st half = t'

$\Rightarrow \dfrac{s}{2} = \dfrac{1}{2}at'^2 \Rightarrow t' = \sqrt{\dfrac{s}{a}}$

$\therefore$ Time taken for covering 2nd half of distance,

$$t_{req} = \sqrt{\frac{2s}{a}} - \sqrt{\frac{s}{a}}$$

$$= \sqrt{\frac{2s}{a}}\left[1 - \frac{1}{\sqrt{2}}\right]$$

$$= t\left[1 - \frac{1}{\sqrt{2}}\right]$$

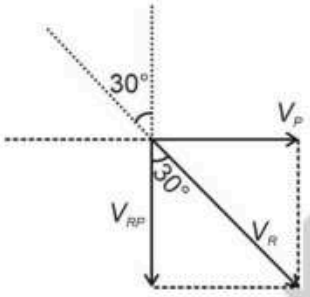
41. Answer (4)

Hint & Sol.: To reach a point directly opposite to the starting point speed of swimmer (in still water) must be more than speed of river, here V_{sr} is less than V_r. So here, this case is not possible.

42. Answer (2)

Hint: $\overrightarrow{V_{R/P}} = \overrightarrow{V_R} - \overrightarrow{V_P}$, draw velocity triangle and find $\left|\overrightarrow{V_{RP}}\right|$.

Sol.: From velocity triangle



$$\tan 30° = \frac{V_P}{V_{RP}}$$

$$\Rightarrow V_{RP} = \frac{V_P}{\tan 30°} = \frac{10}{\frac{1}{\sqrt{3}}}$$

$= 10\sqrt{3}$ km/h

43. Answer (4)

Hint: Write $\vec{u}$ and $\vec{v}$ at time t and use $\vec{u}.\vec{v} = 0$

Sol.: Here let $\vec{u} = (u\cos\theta\hat{i} + u\sin\theta\hat{j})$ and velocity at time t will become

$$\vec{v} = u\cos\theta\hat{i} + (u\sin\theta - gt)\hat{j}$$

Now as given

$\vec{v} \perp \vec{u}$

$\therefore \vec{u}\cdot\vec{v} = 0$

$\Rightarrow u^2\cos^2\theta + u^2\sin^2\theta - u\sin\theta.gt = 0$

$\Rightarrow u^2 - u\sin\theta gt = 0$

$$\Rightarrow t = \frac{u}{g\sin\theta}$$

$$\Rightarrow t = \frac{15}{10 \times 0.5}$$

$= 3$ s

Here, total time of flight of projectile is $T = 1.5$ s

So, no such case is possible.

44. Answer (4)

Hint & Sol.: In uniform circular motion, $\vec{a}_{net}$ is towards centre O like $\overrightarrow{PO}$. In non-uniform circular motion, $\vec{a}_{net}$ may be along $\overrightarrow{PB}$.

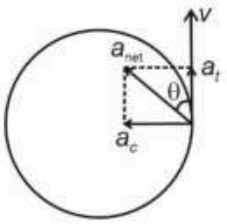
45. Answer (1)

Hint: Draw $\vec{a}$ and $\vec{v}$ at time t

Sol.: $a_t = \frac{d}{dt}|\vec{v}| = \frac{d}{dt}kt = k$

and $a_c = \frac{v^2}{r} = \frac{k^2t^2}{r}$.

From figure



$$\tan\theta = \frac{a_c}{a_t} = \frac{k^2t^2/r}{k}$$

$$\tan\theta = \frac{kt^2}{r}$$

$$\theta = \tan^{-1}\left(\frac{kt^2}{r}\right)$$

[CHEMISTRY]

46. Answer (3)

Hint: Number of molecule = mole × N_A

Sol.:

Initial molecules $= \frac{2.24\times10^{-3}}{22.4}\times N_A = 10^{-4}\ N_A$

Removed molecules $= \frac{1.4\times10^{-3}}{28} N_A$

$= 0.5 \times 10^{-4}\ N_A$

Left molecule $= 10^{-4}\ N_A - 0.5 \times 10^{-4}\ N_A$

$= 0.5 \times 10^{-4}\ N_A$

$= 5 \times 10^{-5} \times 6.02 \times 10^{23}$

$= 3 \times 10^{19}$

47. Answer (4)

Hint: Homogeneous mixture has uniform composition throughout the mixture.

Sol.: Aqueous NaCl, iron filings and air are homogeneous mixtures while milk is a heterogeneous mixture.

48. Answer (1)

Hint: Kilogram is SI unit of mass

Sol.:

- Mole ⇒ Amount of substance
- Kelvin ⇒ Thermodynamic temperature
- Second ⇒ Time

49. Answer (3)

Hint: NaCl does not react with HCl

Sol.: $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

44 g CO_2 is produced from 106 g Na_2CO_3

2.2 g CO_2 will be produced from $= \frac{106}{44}\times 2.2$

$= 5.3$ g Na_2CO_3

Mass of NaCl = 10 – 5.3

= 4.7 g

% NaCl = 47%

50. Answer (1)

Hint: Weighted average $= \frac{\Sigma A_i X_i}{\Sigma X_i}$

$A_i \Rightarrow$ Mass number of ith isotope

$X_i \Rightarrow$ Abundance of ith isotope

Sol.:

Weighted average $= \frac{100\times90 + 99\times8 + 102\times2}{100}$

= 99.96

51. Answer (2)

Hint: For dilution

$M_1V_1 = M_2V_2$

Sol.: Volume of 100 g solution $= \frac{100}{1.8}$ mL

Molarity of conc. $H_2SO_4 = \frac{98\times1.8\times1000}{98\times100} = 18$ M

$18 \times V = 1 \times 0.5 \times 1000$

$V = \frac{500}{18}$

V = 27.78 ml

52. Answer (4)

Hint: Cisplatin is used in cancer treatment.

Sol.:

- 2.00 has three significant figures while 200 has only one.
- Bronze is homogeneous mixture.

53. Answer (4)

Hint: Mole of 0.15 g $CO_3^{2-} = \frac{0.15}{60} = 2.5\times10^{-3}$

Sol.: Number of ions = $2.5 \times 10^{-3} \times 6 \times 10^{23}$

$= 1.5 \times 10^{21}$ ions

Charge = $1.5 \times 10^{21} \times 2 \times 1.6 \times 10^{-19}$

= 480 C

54. Answer (1)

Hint: For minimum molecular mass, given molecule must have one, Ca atom

Sol.: $\%Ca = \frac{1\times40}{M}\times100$

$0.01 = \frac{40}{M}\times100$

$M = 4 \times 10^5$ u

$\therefore$ Molar mass = 4×10^5 g

= 400 kg

55. Answer (2)

Hint: $BaCl_2$ is limiting reagent in given reaction

Sol.: $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

Mole of $Na_2SO_4 = \frac{9.1}{142} = 0.064$

Mole of $BaSO_4 = \frac{11.65}{233} = 0.05$ mol

$BaCl_2$ is limiting reagent

To produce 0.05 mol $BaSO_4 \rightarrow$ 0.05 mol $BaCl_2$

$= 0.05 \times 208$

$= 10.4$ g

56. Answer (2)

Hint: Number of atoms = mole × N_A × atomicity

Sol.: 2.24 L of SO_3 at STP = 0.1 mol

Number of atom = $0.1 \times N_A \times 4$

$= 0.4\ N_A$

4.8 g of ozone = 0.1 mol

Number of atom = $0.1 \times N_A \times 3$

$= 0.3\ N_A$

4.4 g of N_2O = 0.1 mol

Number of atom = $0.1 \times N_A \times 3$

$= 0.3\ N_A$

0.2 mol of $NH_3 = 0.2 \times N_A \times 4$ atoms

$= 0.8\ N_A$

0.2 N_A molecules of water = $0.2 \times N_A \times 3$ atoms

$= 0.6\ N_A$ atoms

57. Answer (1)

Hint and Sol.: According to law of gaseous volume equal volume of gases at given temperature and pressure contains equal number of molecules.

58. Answer (4)

Hint:

Molarity

$$M = \frac{\text{Mass of solute} \times 1000}{\text{Molar mass of solute} \times V_{\text{Solution}}\,(\text{in ml})}$$

Sol.: $M = \dfrac{W_B \times 1000}{M_B \times V_{\text{Solution}}}$

$= \dfrac{3.42 \times 1000}{342 \times 200}$

$= 0.05$ M

59. Answer (4)

Hint: Micro is used for 10^{-6}

Sol.: 10^{-15} – Femto

10^{12} – tera

10^{15} – peta

60. Answer (1)

Hint: $N_2 + 3H_2 \rightarrow 2NH_3$

H_2 will be limiting reagent

Sol.:

	N_2	$+3H_2$	$\rightarrow 2NH_3$
	14 g	2 g	
	0.5 mol	1 mol	0
t = t	0.5–0.33	0	0.66 mol

Mass of NH_3 formed $= \dfrac{2}{3} \times 17$

$= 11.33$ g

Volume of NH_3 produced at STP $= \dfrac{2}{3} \times 22.4$

$= 14.93$ L

Mole of HCl required to neutralise $NH_3 = \dfrac{2}{3}$

as $NH_3 + HCl \rightarrow NH_4Cl$

$1 \times V = \dfrac{2}{3}$

$V = \dfrac{2}{3}$L

$= 666.67$ mL

61. Answer (1)

Hint: On heating water of hydration will be removed which is responsible for mass loss.

Sol.: $Na_2SO_4 \cdot xH_2O \xrightarrow{\Delta} Na_2SO_4 + xH_2O$

Mole of anhydrous $NaSO_4 = \dfrac{\text{Mole of } H_2O}{x}$

$\dfrac{44.1}{142} = \dfrac{55.9}{18 \times x}$

$x = 10$

62. Answer (3)

Hint: Only acidic gas gets absorbed in basic KOH pellets

Sol.:

$$\underset{\substack{t=0\\ t=t}}{\underset{0.1\,\text{mol}}{HCOOH}} \xrightarrow{\text{Conc. } H_2SO_4} \underset{0.1}{\underset{0}{H_2O(\ell)}} + \underset{0.1}{\underset{0}{CO(g)}}$$

$$\underset{\substack{t=0\\ t=t}}{\underset{0.1\,\text{mol}}{H_2C_2O_4}} \xrightarrow{\text{Conc. } H_2SO_4} \underset{0.1}{\underset{0}{H_2O(\ell)}} + \underset{0.1}{\underset{0}{CO(g)}} + \underset{0.1}{\underset{0}{CO_2(g)}}$$

Since CO_2 will get absorbed in KOH, remaining gas product will contain only CO gas

Mole of CO gas = 0.2

Mass of CO gas = 0.2 × 28

= 5.6 g

63. Answer (1)

Hint: 3 L of O_2 gives 2 L of O_3

Sol.:

	$3O_2$	$\rightarrow 2O_3$
	9 ml	0
	9–0.9	0.6

Total volume = 9 – 0.9 + 0.6

= 8.7 ml

64. Answer (3)

Hint: Hydrocarbon gives CO_2 and H_2O on complete combustion.

Sol.: $C_4H_{10} + \frac{13}{2}O_2 \rightarrow 4CO_2 + 5H_2O$

x L C_4H_{10} will give 4x L CO_2

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

(3 – x) L C_3H_8 will give (9 – 3x) L CO_2

Total volume of CO_2 produced = 10 L

4x + 9 – 3x = 10

x = 1 L = volume of C_4H_{10}

3 – x = 2 L = volume of C_3H_8

65. Answer (1)

Hint: % carbon = 100 – (71.7 + 4.04) = 24.26%

Sol.:

	C	H	Cl
	24.26	4.04	71.7
Atomic ratio	$\frac{24.26}{12} = 2.02$	$\frac{4.04}{1} = 4.04$	$\frac{71.7}{35.5} = 2.02$
Simplest ratio	1	2	1

Empirical formula will be CH_2Cl

66. Answer (2)

Hint: Volume of 1 mol gas at STP = 22.4 L

Sol.: Mol of $CO_2 = \frac{11.2}{22.4} = 0.5$ mol

Mass = mol × MW

= 0.5 × 44 = 22 g

67. Answer (4)

Hint: During neutralisation gram equivalent of acid must be equal to that of base.

Sol.: Mass of $CaCO_3$ = x

Mass of Na_2CO_3 = 1.03 – x

$(n_{eq})_{CaCO_3} + (n_{eq})_{Na_2CO_3} = (n_{eq})_{HCl}$

$\frac{x}{50} + \frac{1.03 - x}{53} = 20 \times 10^{-3}$

3x + 51.5 = 53

x = 0.5 g

% $CaCO_3$ = $\frac{0.5 \times 100}{1.03} = 48.54\%$

68. Answer (4)

Hint: Canal rays consist of positively charged particles.

Sol.: e/m ratio of canal rays depends on the nature of gas.

The behaviour of canal rays is opposite to that of cathode rays.

69. Answer (3)

Hint: In order to show photoelectric effect, energy absorbed by metal should be greater than its work function.

Sol.: $E(eV) = \frac{1240}{\lambda(nm)}$

$E(eV) = \frac{1240}{248} = 5$ eV

70. Answer (1)

Hint: $\tilde{\nu} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$ cm^{-1}

R = 109677 cm^{-1}

Sol.: Shortest wavelength means that the transition would belong to maximum energy.

$\tilde{\nu} = R\left[\frac{1}{9} - \frac{1}{\infty}\right]$

$= \frac{109677}{9} = 12186.3$ cm^{-1}

71. Answer (4)

Hint: $E_n = -\frac{13.6\ Z^2}{n^2}$ eV

Z for He^+ ion is 2

n = 2, 3, 4, 5, etc

Sol.: $E_2 = -\frac{13.6 \times 4}{4} = -13.6$ eV

$E_3 = -\frac{13.6 \times 4}{9} = -6.044$ eV

$E_4 = -\frac{13.6 \times 4}{16} = -3.4$ eV

72. Answer (4)

Hint: $\Delta x \times \Delta V \geq \frac{h}{4\pi m}$

Sol.: $\Delta V = \frac{50 \times 3}{100} = 1.5$ m/s

$\Delta x = \frac{6.626 \times 10^{-34}}{4 \times 3.14 \times 30 \times 10^{-3} \times 1.5}$

= 0.0117 × 10^{-31} m

= 1.17 × 10^{-33} m

73. Answer (3)

Hint: $\lambda = \frac{h}{\sqrt{2m\,K.E.}}$

Sol.: $\lambda \propto \frac{1}{\sqrt{m\,K.E.}}$

$(K.E.)_{He} = 4\,(K.E.)_{Ne}$

$\frac{\lambda_{Ne}}{\lambda_{He}} = \frac{\sqrt{(K.E.)_{He} \times m_{He}}}{\sqrt{(K.E.)_{Ne} \times m_{Ne}}} = \frac{\sqrt{4 \times 4}}{\sqrt{1 \times 20}}$

$= \frac{4}{2\sqrt{5}}$

$= 2 : \sqrt{5}$

74. Answer (2)

Hint and Sol.: n, ℓ and m_ℓ arise as a consequence in the solution of Schrodinger equation.

75. Answer (4)

Hint: The orbitals are filled in increasing order of energy according to Aufbau principle.

Sol.: The given electronic configuration is violating all three rules

a. Pauli exclusion principle

b. Hund's rule of maximum multiplicity

c. Aufbau principle

76. Answer (2)

Hint: $3d_{z^2}$ has zero nodal plane.

Sol.:

- $3d_{z^2}$: Zero nodal plane, zero radial node
- $2p_z$: xy nodal plane, zero radial node
- $2s$: Zero nodal plane, one radial node
- $4d_{xy}$: yz and xz nodal plane, 1 radial node

77. Answer (1)

Hint: Species that have same number of neutrons are isotones

Sol.: $^{36}_{16}S$ and $^{39}_{19}K$ have 20 neutrons, so, are isotones to each other.

78. Answer (4)

Hint: According to Bohr's model of atom, radius of nth orbit $\propto \frac{n^2}{Z}$

Sol.:

- $r_n \propto \frac{n^2}{Z}$
- $|E_n| \propto \frac{Z^2}{n^2}$
- $V_n \propto \frac{Z}{n}$
- $mvr = \frac{nh}{2\pi}$

79. Answer (1)

Hint: The total number of electrons in any given subshell is $4\ell + 2$

Sol.: $\ell = 4$ for 5 g

Number of electrons = 4 × 4 + 2 = 18

80. Answer (2)

Hint: Energies of the orbitals in the same subshell decrease with increase in the atomic number.

Sol.: The correct order of energy is

$E_{2s}(H) > E_{2s}(Li) > E_{2s}(Na) > E_{2s}(K)$

81. Answer (2)

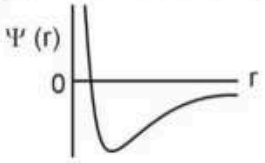
Hint: Given element is Pd (Z = 46)

Sol.: The correct outermost electronic configuration is $4d^{10}\ 5s^0$.

82. Answer (2)

Hint: For 2s orbital, number of nodes = 1

Sol.: For 2s orbital $\Psi(r)$ is plotted against r as



83. Answer (3)

Hint: $E = \frac{hc}{\lambda}$

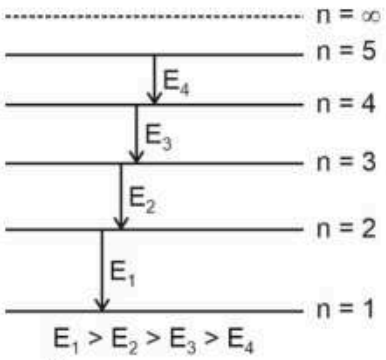
Sol.: $E \propto \frac{1}{\lambda}$

$\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = \frac{900}{300} = 3 : 1$

84. Answer (1)

Hint: $\Delta E = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

Sol.:



85. Answer (3)

Hint: $\Delta E = \frac{hc}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

Sol.: At STP conditions,

Number of moles of H_2 gas $= \frac{1.12}{22.4} = 0.05$

Energy required to dissociate given sample into gaseous atoms = 0.05 × 436 = 21.8 kJ

Second excited state = $n_2 = 3$, $n_1 = 1$

$$\Delta E = 2.18 \times 10^{-18}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) J/atom$$

$$= 2.18 \times 10^{-18}\left(1 - \frac{1}{9}\right)$$

$$= 2.18 \times 10^{-18}\left(\frac{8}{9}\right)$$

$= 1.937 \times 10^{-18}$ J/atom

To excite 0.05 mol of H atoms

$= 0.05 \times 6.02 \times 10^{23} \times 1.937 \times 10^{-18} \times 2$

$$= \frac{0.583 \times 2 \times 10^5}{10^3} kJ$$

= 58.3 × 2 = 116.6 kJ

Total energy required = 138.4 kJ

86. Answer (1)

Hint: Orbital angular momentum $\mu_l = \sqrt{\ell(\ell+1)}\frac{h}{2\pi}$

Sol.:

a. For s subshell, $\mu_l = \sqrt{0(0+1)}\frac{h}{2\pi} = 0$

b. For f subshell, $\mu_l = \sqrt{3(3+1)}\frac{h}{2\pi} = \sqrt{12}\frac{h}{2\pi}$

c. For d subshell, $\mu_l = \sqrt{2(2+1)}\frac{h}{2\pi} = \sqrt{6}\frac{h}{2\pi}$

d. For p subshell, $\mu_l = \sqrt{1(1+1)}\frac{h}{2\pi} = \sqrt{2}\frac{h}{2\pi}$

87. Answer (2)

Hint: The outermost electronic configuration of Cr^{3+} is $3d^3$

Sol.: The outermost electronic configuration is as follows

Fe^{2+}, $3d^6$ ↑↓ | ↑ | ↑ | ↑ | ↑ 4 unpaired electrons

Cr^{3+}, $3d^3$ ↑ | ↑ | ↑ | | 3 unpaired electrons

Mn^{2+}, $3d^5$, 5 unpaired electrons

Co^{2+}, $3d^7$, 3 unpaired electrons

88. Answer (2)

Hint: $h\nu = h\nu_0$ + K.E.

Sol.: With increase in frequency of incident light, there will be increase in kinetic energy of ejected electron.

89. Answer (1)

Hint: 4th excited state is $n_2 = 5$

Sol.: $n_2 = 5$ to $n_1 = 1$

Number of lines in visible region = 5 – 2 = 3

90. Answer (3)

Hint and Sol.: If the value of ℓ is 1, the allowed values of m_ℓ are –1, 0, +1

[BOTANY]

91. Answer (2)

Hint: Carbohydrates or sugars are attached to lipids and act as receptors.

Sol.: The mammalian RBCs, is composed of 52% proteins and 40% lipids.

92. Answer (2)

Hint: Lysosomes show polymorphism.

Sol.: ER may or may not have ribosomes ("organelle within organelle").

Golgi bodies secrete lubricating mucilage for root tip.

Lysosomes show polymorphism.

93. Answer (1)

Hint: Mitochondria and chloroplast are semi-autonomous organelles.

Sol.: Mitochondria and chloroplast support the hypothesis that both originated as independent free-living structures.

94. Answer (3)

Hint: Centrosome is involved in spindle apparatus formation.

Sol.: Lysosomes are involved in digesting non-functional cells or organelles.

Microsomes are formed due to cell disruption, primarily from fragments of the ER.

95. Answer (2)

Hint: The larger subunit contains 5S and 5.8S rRNA.

Sol.: 16S rRNA is present in smaller subunit of 70S ribosomes.

96. Answer (2)

Hint: Cell is the basic structural and functional unit of life.

Sol.: All living organisms are composed of cells and products of cells and each cell arises from a pre-existing cell.

97. Answer (1)

Hint: Growth is a characteristic of living organisms.

Sol.: Growth and reproduction are characteristic features of living beings.

98. Answer (1)

Hint: Daughter cell after mitosis is genetically similar to each other.

Sol.: Meiosis II occurs in a haploid cell with same DNA content as in parent cell in G_1 phase.

99. Answer (3)

Hint: Congression is observed during metaphase.

Sol.: Chromosome identity is lost in telophase.

100. Answer (2)

Hint: It will inhibit the stage which begin when complete nuclear envelope disappears.

Sol.: Metaphase stage will be immediately inhibited by the activity of colchicine.

101. Answer (4)

Hint: Mitotic phase follows G_2 phase.

Sol.: In S phase, DNA replication and histone synthesis occurs

102. Answer (4)

Hint: Meiosis is also termed as reductional division.

Sol.: Meiosis maintains the number of chromosomes during sexual reproduction.

103. Answer (2)

Hint: Kingdom have less number of similar characters than lower taxa.

Sol.: The number of similar characteristics goes on decreasing from species to kingdom.

104. Answer (3)

Hint: Phylum is used for animals.

Sol.: Dicotyledonae is a class.

105. Answer (3)

Hint: Both genus and species are italicized.

Sol.: *Musca domestica* is correctly printed form of binomial name of housefly.

106. Answer (2)

Hint: Cell wall is made up of cellulose.

Sol.: Algae have cell wall made of cellulose, galactans, mannans and minerals like calcium carbonate.

107. Answer (3)

Hint: Ribosome is not part of endomembrane system.

Sol.: Golgi bodies, are involved in packaging of materials. Numerous RER is seen in cells involved in protein synthesis.

108. Answer (2)

Hint: Label (c) is radial spoke.

Sol.: Interdoublet bridge is labelled as (d) and they connect the peripheral microtubules to each other. Plasma membrane of RBC is having approximately 52% protein and 40% lipids.

109. Answer (3)

Hint: Reserve material in prokaryotic cells are stored in the cytoplasm in the form of inclusion bodies.

Sol.: Gas vacuoles are not meant for storage of reserve material; they provide buoyancy to the cell. Phosphate granules, cyanophycean granules and glycogen granules are inclusion bodies.

110. Answer (4)

Hint: Centrosome is absent in plant cell and present in animal cell.

Sol.: Centrioles have nine peripheral triplet fibrils.

111. Answer (2)

Hint: Antonie Von Leeuwenhoek first saw and described the live cell.

Sol.: George Palade — discovered ribosome, a dense particle composed of RNA and proteins.

Camillo Golgi — discovered densely stained reticular structures near the nucleus.

Robert Brown — first described the nucleus.

112. Answer (3)

Hint: 80S ribosome is observed in the cytosol of eukaryotes.

Sol.: 60S and 40S subunits form 80S ribosome in eukaryotes.

113 Answer (4)

Hint: Centromere have disc shaped structures that bind to spindle fibres.

Sol.: Centromere is the primary constriction of chromosome. Kinetochore is a disc-shaped structure present on centromere that binds spindle fibres.

114. Answer (2)

Hint: Interphase nucleus does not have twice the number of chromosomes than the nucleus of mitosis.

Sol.: The interphase nucleus has highly extended and elaborate nucleoprotein fibres.

115. Answer (3)

Hint: Lipoproteins and glycogen synthesis is helped by endomembrane system.

Sol.: ER helps in the synthesis of proteins, lipoproteins and glycogen.

116. Answer (4)

Hint: A special membranous structure formed by the extensions of plasma membrane into the cell performs oxidative phosphorylation.

Sol.: Mesosome helps in respiration in prokaryotes.

117. Answer (2)

Hint: It is the organelle that contains chromatin.

Sol.: Nuclear outer membrane usually remains continuous with the endoplasmic reticulum and also bears ribosomes on it.

118. Answer (1)

Hint: Metaphase I brings variation in gametes.

Sol.: The stage 'b' is Metaphase II and it will undergo further stage by APC.

The stage 'b' does not bring variation in gametes.

119. Answer (3)

Hint: The compaction of chromosomes continues throughout the first stage of karyokinesis.

Sol.: Bivalent or a tetrad is more clearly visible in pachytene stage which is relatively long-lived than leptotene and zygotene.

120. Answer (2)

Hint: Diplotene is the second last sub-stage of prophase I.

Sol.: The beginning of diplotene is recognised by the dissolution of the synaptonemal complex and the tendency of the recombined homologs of bivalents to separate from each other except at the sites of crossover.

121. Answer (2)

Hint: G_1 phase is followed by S phase.

Sol.: The correct sequence is:

$f(G_1) \rightarrow c(S\ phase) \rightarrow b(Early\ Metaphase) \rightarrow e(Metaphase) \rightarrow d(Anaphase)$

122. Answer (1)

Hint: The amount of DNA is doubled after S phase.

Sol.: Onion cell in metaphase will have 16 chromosomes and double amount of DNA (4C).

123. Answer (3)

Hint: The final stage of karyokinesis is telophase.

Sol.: At telophase, chromosomes that have reached their respective poles decondense and lose their individuality.

124. Answer (1)

Hint: The final stage of M-phase is cytokinesis.

Sol.: The cytokinesis in animal cell starts with the appearance of a furrow in the plasma membrane. The furrow gradually deepens and ultimately joins in the centre dividing the cell cytoplasm into two.

125. Answer (2)

Hint: The simple precursor is the cell-plate that represents the middle lamella.

Sol.: Middle lamellae is a layer mainly of calcium pectate which holds or glues the different neighbouring cells together.

126. Answer (1)

Hint: The liquid endosperm in coconut is a syncytium.

Sol.: Here, karyokinesis is not followed by cytokinesis. As a result, a multinucleate condition arises, leading to the formation of a syncytium (e.g., liquid endosperm in coconut)

127. Answer (3)

Hint: Crossing over occurs between homologous chromosomes.

Sol.: Crossing over occurs between non-sister chromatids of homologous chromosomes.

128. Answer (2)

Hint: It leads to the separation of chromosomes.

Sol.: Shortening of microtubules attached to kinetochores occurs during Anaphase II.

129. Answer (1)

Hint: Meiosis is called as reductional division.
Sol.: Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing higher plants. As it reduces the number of chromosomes by half resulting in production of haploid gametes.

130. Answer (1)

Hint: Mitotic division occurs in the continuously dividing cell.

Sol.: Mitotic division occurs in the root apical meristem, lateral cambium and embryo.

131. Answer (3)

Hint & Sol.: Yeast cell progress through the cell cycle in only about 90 minutes.

132. Answer (1)

Hint: Thylakoids have the chlorophyll pigments.

Sol.: Cytoskeleton is made of microtubules, microfilaments and intermediate filaments.

133. Answer (3)

Hint: Sapindales is taxa at higher level than family.

Sol.: Sapindales is an order.

134. Answer (2)

Hint: Order includes related families.

Sol.: Family: Has a group of related genera with still less number of similarities as compared to genus and species.

Genus: Comprises a group of related species.

135. Answer (1)

Hint: Carnivora is an order.

Sol.: Felidae and Canidae are families that are included in order Carnivora.

[ZOOLOGY]

136. Answer (2)

Hint: Only one H-atom is present in its structure

Sol.: To perform chemical analysis, the living tissue is grinded in trichloroacetic acid (Cl_3CCOOH) using a mortar and a pestle.

$ClCH_2COOH$ represents chloroacetic acid.

137. Answer (3)

Hint: Polymer of fructose

Sol.: Glycogen, starch and cellulose are polymers of glucose, while inulin is a polymer of fructose.

138. Answer (1)

Hint: Hb exhibits quaternary structure.

Sol.: Physicists conjure up the three-dimensional views of molecular structures, while biologists describe the protein structure at four levels.

An α-helix contains intramolecular hydrogen bonding. This specific hydrogen bond forms between the carbonyl oxygen of one amino acid residue and the amide hydrogen of an amino acid located four residues farther down the chain.

139. Answer (4)

Hint: Hollow organs like bronchioles and oviduct

Sol.: Cilia beat in a synchronized directional manner to move ova through the fallopian tubes and clear mucus out of the respiratory bronchioles. Zygote is formed in the fallopian tube.

Compound epithelium is present in the inner lining of ducts of salivary glands while simple cuboidal epithelium lines tubular parts of nephrons.

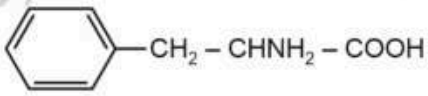
140. Answer (3)

Hint: Pavement epithelium

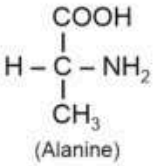
Sol.: Simple squamous epithelium is composed of a single thin layer of flattened cells with irregular boundaries which makes it perfect for diffusion or filtration.

141. Answer (1)

Hint: Eliminate the sulfur containing amino acids.



(Phenylalanine)



(Alanine)

Cysteine and methionine have a linear structure. They are sulphur containing amino acids.

142. Answer (2)

Hint: They are not the structural units of the neural system.

Sol.: Neuroglia constitute the non-conducting supportive part that accounts for more than fifty per cent of the total volume of the nervous tissue.

143. Answer (3)

Hint: Disulphide bonds are hydrolysed

Sol.: High temperature causes irreversible denaturation due to disruption of the tertiary structure of the protein.

Enzymes are mostly proteins.

144. Answer (4)

Hint: Glycogen is stored in the liver.

Sol.:

Maltose	–	Disaccharide composed of glucose units
Glycogen	–	Branched storage polysaccharide
Cellulose	–	Absence of complex helices
Chitin	–	Polymer of N-acetylglucosamine

145. Answer (2)

Hint: A nucleic acid

Sol.: DNA polymerases are proteins. Ribozymes are catalytic RNA molecules that can actively catalyse certain chemical reactions.

Nucleases are hydrolases. They are also proteinaceous enzymes.

146. Answer (4)

Hint: Carbohydrates are polysaccharides.

Sol.: Trypsin, insulin, GLUT-4 and collagen are proteins. Proteins are always heteropolymers.

147. Answer (3)

Hint: Activation energy

Sol.: Enzymes function by lowering the activation energy barrier of the reaction needed to reach the transient unstable state.

148. Answer (2)

Hint: Osteoblasts are bone forming cells.

Sol.: Osteocytes are mature bone cells trapped inside the fluid-filled spaces of the calcified matrix, which are called lacunae.

The osteon consists of a central canal (called the Haversian canal), which is surrounded by concentric rings (lamellae) of matrix.

149. Answer (4)

Hint: Stratified in nature

Sol.: Compound epithelium is composed of multiple layers, making it an excellent protective barrier, but it has a limited role in secretion and absorption.

150. Answer (3)

Hint: Most abundant among all the connective tissues

Sol.: Areolar tissue is a kind of loose connective tissue. It is located beneath the skin, acting as a supportive framework for the epithelium. It also houses fibroblasts, mast cells, *etc*.

Tendons and ligaments are dense regular connective tissues.

151. Answer (2)

Hint: Ligaments and tendons

Sol.: In the dense regular connective tissues, the collagen fibres are present in rows between many parallel bundles of fibres.

Dense irregular connective tissue has fibroblasts and many fibres (mostly collagen) that are oriented differently.

Dense regular connective tissues are poorly vascularized.

152. Answer (3)

Hint: Opposite strands are joined by H-bonds

Sol.: Uracil is present in RNA, while it is absent in DNA. Phosphodiester bonds join adjacent nucleotides of the same strand.

Adenine and guanine are purines.

153. Answer (2)

Hint: Peptide chain can have variable lengths.

Sol.: Only serine has $-CH_2OH$ group as the 'R'-group.

Number of peptide bonds presents in a polypeptide chain is equal to = [Number of amino acids – 1]

An oligopeptide is a short chain of roughly 2 to 20 amino acids, while a polypeptide is a much longer chain containing more than 20 amino acids.

154. Answer (3)

Hint: Odd number

Sol.: Morphine, antibiotics and spices are secondary metabolites. Many secondary metabolites are useful to human. Amino acids and lecithin are primary metabolites.

155. Answer (4)

Hint: Palmitic acid belongs to this group.

Sol.: Proteins, polysaccharides and nucleic acids have molecular weights in the range of 10 thousand daltons and above. The molecular weights of lipids do not exceed 800 Da.

Proteins, polysaccharides and nucleic acids are called biomacromolecules.

156. Answer (2)

Hint: Think about the structure of cholesterol

Sol.: Living organisms have a number of carbon compounds in which heterocyclic rings can be found. When a pentose sugar is found attached to a nitrogenous base, it is called a nucleoside.

Cholesterol contains homocyclic rings.

157. Answer (3)

Hint: WBCs are present

Sol.: Blood is a fluid connective tissue. It contains neutrophils, monocytes, lymphocytes, *etc.* These cells provide immunity.

Adipose tissue has a semi-fluid ground substance.

The intercellular material of cartilage is solid and pliable and resists compression.

Bones have a hard and non-pliable ground substance rich in calcium salts and collagen fibres, which gives bone its strength.

158. Answer (2)

Hint: Unstriated muscle fibres

Sol.: The features mentioned in the question hold true for smooth muscle fibres.

Smooth muscle fibres are involuntary and unbranched.

They contain actin and myosin.

159. Answer (3)

Hint: A type of connective tissue

Sol.: The mentioned features are of areolar connective tissue, which acts as a packing material.

Nervous and muscle tissues are involved in conduction of impulses.

Loosely arranged fibres are found in the ground substance of loose connective tissue.

Bones store minerals.

160. Answer (2)

Hint: Other than co-factors

Sol.: The apoenzyme is exclusively the protein portion of a conjugated enzyme, which is catalytically inactive until it binds to its specific non-protein co-factor.

Prosthetic groups and co-enzymes are co-factors.

Apoenzyme + Co-factor = Holoenzyme.

161. Answer (4)

Hint: Coordination bond formation

Sol.: Metal ions function by establishing coordination bonds that anchors the active site residues to the substrate.

162. Answer (2)

Hint: Allow rapid transfer of ions

Sol.: Tight junctions help to stop substances from leaking across the tissue. Adhering junctions perform cementing to keep the neighbouring cells together. Intercalated discs are made up of communication junctions and desmosomes.

163. Answer (2)

Hint: Non-pliable matrix

Sol.: Bone is a skeletal connective tissue. Its cells are arranged in concentric layers. Its matrix is hard and non-pliable as it is highly mineralized. Bone is surrounded by a tough sheath, called periosteum.

Cartilage lacks canaliculi.

Cartilage is surrounded by a firm sheath, called perichondrium.

164. Answer (2)

Hint: Some are useful to human

Sol.: While primary metabolites have identifiable functions, we do not at the moment understand the roles or functions of all the secondary metabolites in host organisms.

165. Answer (3)

Hint: Ribose sugar is present

Sol.: The figure represents a nucleotide named guanylic acid. Because it contains ribose sugar, it is the component of RNA (not DNA). Glycosidic bond is present between nitrogenous base (here it is guanine) and sugar. Guanine is a double-ringed structure.

166. Answer (2)

Hint: Move particles in a specific direction

Sol.: Ciliated epithelium is present in the respiratory tract.

Transitional epithelium is present in the urinary bladder. Simple squamous epithelium is present in the lining of alveoli and intestine.

167. Answer (3)

Hint: Modified epithelium

Sol.: Based on cell number, glands are of two types:

(1) Unicellular glands – Goblet cells of the alimentary canal; they consist of isolated single cells.

(2) Multicellular glands – Salivary glands, sweat gland, *etc.*; glands that are made up of cluster of cells.

168. Answer (3)

Hint: Cylindrical in shape

Sol.: Cardiac muscle fibres are characterised by the presence of branched, cylindrical and uninucleated condition. These fibres show presence of intercalated discs.

169. Answer (2)

Hint: Both are connective tissues

Sol.: Ligaments are dense regular connective tissues, while cartilage is specialized (skeletal) connective tissue. Both have matrix and collagen fibres. Osteocytes are present in bone.

Cells of cartilage are called chondrocytes.

170. Answer (1)

Hint: Tertiary structure possesses disulfide bonds.

Sol.: Tertiary structure is absolutely necessary for the many biological activities of proteins. Denaturation destroys tertiary structure without breaking peptide bonds.

Proteins exhibit a wide range of biological functions because of differences in the sequence of amino acids.

171. Answer (4)

Hint: Active sites.

Sol.: Enzymes are substrate specific.

A lower K_m value means high affinity, while a higher K_m value means low affinity.

172. Answer (3)

Hint: True for lecithin

Sol.: Bilayer formation is characteristic of phospholipids. Palmitic acid and cholesterol do not contain phosphate.

Glycerol is not present in polysaccharides.

173. Answer (3)

Hint: Cell wall of plant is made up of cellulose.

Sol.: Humans cannot digest cellulose due to the lack of cellulase enzyme.

Cellulose is a structural polysaccharide, present in the plant cell wall.

174. Answer (4)

Hint: Biceps

Sol.: All animals do not have tissue level of body organization.

In nearly all epithelial tissues, cell junctions provide both structural and functional links between individual cells.

175. Answer (4)

Hint: Ligaments connect bone to bone

Sol.: Both tendons and ligaments are dense regular connective tissues. Tendons connect muscle to bones while ligaments connect bone to bone.

Tendons are inelastic in nature.

176. Answer (3)

Hint: Present in ear pinnae

Sol.: Cartilage is a type of skeletal connective tissue. Its matrix is rich in chondroitin sulphate. It is avascular.

Blood and lymph are fluid connective tissues.

177. Answer (3)

Hint: Polysaccharides are polymers

Sol.: Lipids contain oxygen in their structures. Lipids are always non-polymeric.

Lipids are more reduced than carbohydrates because they contain a higher ratio of carbon-hydrogen bonds and very little oxygen.

Carbohydrates can exist in both open chain (acyclic) and ring (cyclic) structures.

178. Answer (2)

Hint: Carbon is the second highest element present in humans.

Sol.: Hydrogen - 9.5%

Carbon - 18.5%

Oxygen - 65%

Nitrogen - 3.3%

179. Answer (3)

Hint: Myelin sheath

Sol.: The nervous system relies heavily on complex structural lipids to build insulating myelin sheaths and create precise membrane spaces for rapid electrical signalling.

180. Answer (2)

Hint: Responsible for metabolic activity in neurons

Sol.: The axon hillock is the cone-shaped region of the cell body from which the axon originates. Dendrites receive nerve impulses. The cell body contains cell organelles.

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